

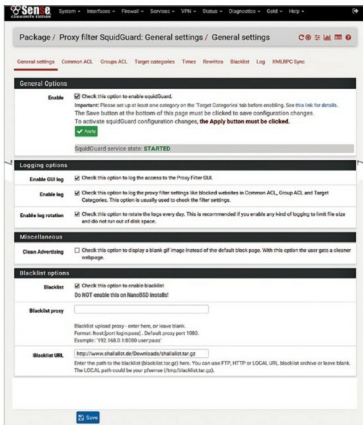


Figure 6: SquidGuard configuration

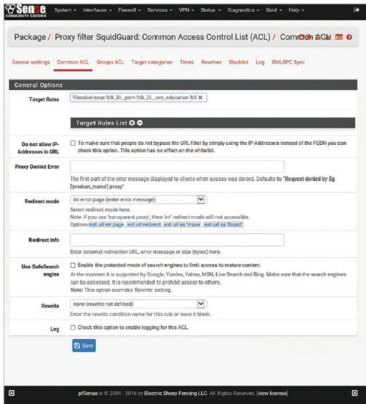


Figure 7: SquidGuard common ACL

further installation and configuration.

Let us take a look at several interesting pfSense features and packages.

A selection of pfSense's inherent features

- Creating ACLs to control access from the internal IP addresses towards the external IP addresses for the required ports (services)
- Network address translation
- DHCP server
- NTP server
- ClamAV anti-virus

pfSense packages

Various packages are available for online installation from *System – Package Manager – Available Packages* menu. This menu lists all such available packages including:

- Iftop – A real-time interface monitor
 - Nmap – A network exploration and security utility
 - Openvpn client export – This exports pre-configured openvpn client settings for Windows and MAC
 - OpenBGPD, Routed and Quagga_OSPF – Offers support for BGP, RIP V1/V2 and OSPF routing protocols
 - Snort and Suricata – For intrusion detection and prevention
 - Squid – This is a high performance Web proxy cache, which also supports SSL filtering
 - SquidGuard – This is a Web proxy URL filter
- Out of these, the most popular is Squid for caching and SquidGuard for website filtering.

Installation and configuration of the Squid proxy server

Go to **System – Package Manager – Available Packages** and click on the *Install* button corresponding to Squid.

The Squid proxy will download and install automatically. 'Success' at the end of the installation screen will indicate that the Squid installation is complete. This will add two menu items under *Services* – *Squid Proxy Server* and *Squid Reverse Proxy*.

Start with the **Services – Squid Proxy Server** menu. Figure 5 elaborates on the required configuration parameters.

One by one, browse through the following sub-menus — *Local Cache*, *ACLs*, *Traffic Mgmt*, *Authentication*, *Users* — and make the desired changes or save their default settings.